

MY-VID: An Open-Source Malaysian Vehicle Image Dataset for Intelligent Transportation System and Road Safety

Zi Xuan Lim, Jiehan Teoh and Kian Meng Yap

Abstract— This study proposes MY-VID (MalaYsia Vehicle Image Dataset), a novel open-source vehicle dataset tailored to Malaysian road conditions emphasizing on commercial and private vehicles classification. Motivated by high road accident fatality rates and the absence of localized datasets in Southeast Asia (SEA), this study introduces a privacy-compliant annotated image dataset captured from Malaysian highway. All stages from data collection, anonymization, preprocessing and augmentation were conducted offline to prioritize data security and privacy. These privacy safeguards allowed the dataset to be safely released as an open-source resource on the Zenodo platform for public accessibility. Benchmark experiments with YOLO11 and YOLO12 demonstrate strong performance and highlight the importance of localized dataset in vehicle detection accuracy. The dataset and training framework aim to support future research in intelligent transportation system (ITS), traffic safety analysis and road infrastructure development in Malaysia.

I. INTRODUCTION

Malaysia contributes to face alarming road accident fatalities especially those involving commercial vehicles such as lorries and buses. Despite the proliferation of AI-assisted traffic solutions globally, there remains a lack of open-source vehicle image datasets specific to Malaysian context. In addition, most existing datasets are collected in Western environments with moderate sunlight and different vehicle dimensions which makes them less suitable for deployment for equatorial regions [1]. The SEA context particularly Malaysia features unique environmental challenges such as intense sunlight, glare, tropical rainfall and vehicle density shaped by local road usage and anthropometric preferences [2]. Therefore, this study bridges the gap by presenting a localized privacy-compliant vehicle dataset to support computer vision research and policy efforts to reduce road accident fatalities in Malaysia.

II. DATASET CONSTRUCTION AND TRAINING WORKFLOW

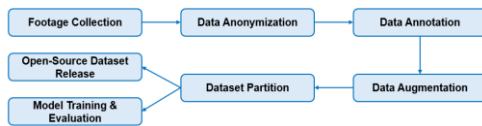


Fig.1. MY-VID dataset construction workflow

Video footage was collected from an urban Malaysian highway, namely Shah Alam Expressway (KESAS) due to its high vehicle throughput and ability to capture diverse vehicle

types. The recordings were taken across different times of the day, on weekdays, weekends and public holidays, under varying traffic densities and weather conditions to ensure the coverage of real-world situations. In compliance with Malaysia Personal Data Protection Act (PDPA), blurring techniques were applied to anonymize identifiable features apart from the focused lanes. Frames were then extracted and annotated using LabelMe annotation tool with a focus on differentiating private and commercial vehicles exported in TXT format. Data augmentation such as flipping and grayscale transformations were applied to increase dataset diversity, followed by splitting into training (70%), validation (20%) and testing (10%) formatted for YOLO model. As a result, 8,832 images with 47,975 vehicles were generated and benchmarked using YOLO11 and YOLO12 models trained on NVIDIA A100 Tensor Core GPU.

III. EXPERIMENTAL RESULTS AND DISCUSSIONS

Based on the result, YOLO11 outperforms the YOLO12 in detecting both commercial and private vehicles across all evaluation metrics. The results suggest YOLO11 is more generalizable for Malaysian traffic conditions in identifying commercial vehicles.

TABLE I. CLASS-WISE DETECTION PERFORMANCE

Class	Model	mAP0.5	mAP0.5:0.95	Precision	Recall	F1 Score
Commercial Vehicle	YOLO 11	0.948	0.797	0.914	0.919	0.916
	YOLO 12	0.942	0.769	0.905	0.907	0.906
Private Vehicle	YOLO 11	0.965	0.781	0.953	0.898	0.925
	YOLO 12	0.949	0.757	0.948	0.858	0.901

IV. CONCLUSION

This study introduces MY-VID, a novel Malaysian open-source vehicle detection dataset built with privacy and accessibility in mind. It not only fills the gap in SEA but also offers a modular framework for supporting any potential academic, governmental and industrial applications to improve Malaysian road safety. For more information, the MY-VID dataset is released and openly available on Zenodo at <https://doi.org/10.5281/zenodo.16866509>.

REFERENCES

- [1] H. Lin *et al.*, “A Study on Data Selection for Object Detection in Various Lighting Conditions for Autonomous Vehicles,” *Journal of Imaging* 2024, Vol. 10, Page 153, vol. 10, no. 7, p. 153, Jun. 2024, doi: 10.3390/JIMAGING10070153.
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